Software Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #62

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Section: 5F

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Donor Partner to post a surplus food batch listing that includes item description, quantity, dietary tags, and a perishable expiry countdown timer.	High	Pass: When the expiry timer reaches zero, the batch automatically transitions to "Expired" and is removed from the claim board within 60 seconds. Fail: An expired batch (timer = 0) remains active/visible on the claim board.	Directly required by the assigned problem statement (FR-001); prevents shelters from claiming unsafe or expired food.
FR-002	Functional	The system shall allow a Shelter Coordinator to browse and filter active surplus food batches by dietary tag, distance, and remaining time-to-expiry.	High	Pass: Applying a dietary-tag filter (e.g., "vegetarian") returns only matching, active batches within 2 seconds. Fail: Filtered results include non-matching batches, or the response takes longer than 2 seconds.	Enables Shelter Coordinators to quickly locate suitable, safe batches, supporting the browse/claim workflow described in the problem statement.
FR-003	Functional	The system shall allow a verified Shelter Coordinator to reserve an active, unexpired food batch for a specific available pickup window.	High	Pass: Reserving a batch immediately decrements its available quantity and marks it "Reserved" for all other shelters within 1 second. Fail: Two shelters can simultaneously reserve the same single-unit batch (double-booking).	Core transaction of the platform; prevents duplicate claims on the same surplus batch.
FR-004	Functional	The system shall generate a unique pickup verification code and record the selected pickup window immediately after a reservation is confirmed.	High	Pass: Every confirmed reservation has exactly one associated verification code, shown to the Shelter Coordinator on the confirmation screen. Fail: A confirmed reservation exists with no code, or two reservations share the same code.	Provides the mechanism used to authenticate the physical handoff of food, directly supporting FR-001's stated pass condition.
FR-005	Functional	The system shall mark a reservation as "Completed" only when the Donor Partner (or authorized staff) validates the Shelter Coordinator's verification code at the pickup location.	Medium	Pass: Entering the correct, unused verification code within the scheduled pickup window updates the reservation status to "Completed" and logs a timestamp. Fail: An incorrect or already-used code is accepted, or a reservation is marked "Completed" without a code check.	Closes the redistribution loop and produces an auditable record for food-safety and accountability purposes.
NFR-001	Nonfunctional	The platform shall dispatch instant push notifications to all verified shelters within a 5 km radius whenever a high-quantity (≥ configured threshold) perishable food batch is listed.	High	Pass: Under simulated peak load (500 concurrent listings/hour), 95% of eligible shelters receive the push notification within 10 seconds of posting, verified via server logs. Fail: More than 5% of notifications exceed 10 seconds latency, or notifications reach shelters outside the 5 km radius.	Directly required by the assigned problem statement (NFR-001); timely alerts are critical given the perishability of the food.
NFR-002	Nonfunctional	The system shall restrict all food-batch posting, reservation, and verification actions to authenticated and verified	High	Pass: Penetration testing confirms unauthenticated API requests to post, reserve, or verify a batch are rejected (401/403) in 100% of test	Protects donor and shelter personal data and prevents fraudulent postings or

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
		accounts, and shall encrypt all stored credentials and personal contact data.		cases, and a database export shows no plaintext passwords or phone numbers. Fail: Any unauthenticated request succeeds, or any credential/contact field is stored in plaintext.	reservations that would undermine trust and food safety.

2. Requirement Quality Review

Functional Requirements (FR-001 – FR-005)

- Clarity: Each FR uses a single, unambiguous "The system shall..." statement naming one actor and one action (e.g., FR-003 names only the Shelter Coordinator reserving one batch for one window), avoiding compound or vague behaviour.
- Testability: Every FR's acceptance criterion states a concrete, observable Pass condition and a concrete Fail condition (e.g., FR-004's code uniqueness can be checked directly against the database), so a tester can assign a binary pass/fail outcome without interpretation.
- Relevance: Each FR maps to exactly one stage of the donation-to-pickup lifecycle described in the problem statement (post → browse → reserve → verify), so no requirement introduces functionality outside the assigned scenario.

Nonfunctional Requirements (NFR-001 – NFR-002)

- Clarity: Both NFRs specify a measurable quality attribute (latency/radius for NFR-001, authentication/encryption for NFR-002) rather than a subjective quality like "secure" or "fast".
- Testability: NFR-001 is testable via load-test logs against a 10-second/5 km threshold; NFR-002 is testable via penetration testing and a database export check — both yield an objective pass/fail result.
- Relevance: NFR-001 is taken directly from the problem statement's perishability constraint; NFR-002 protects the personal and location data that the platform must collect to satisfy FR-001–FR-005, so both are intrinsic to this system rather than generic checklist items.

3. Requirement-to-Use-Case Traceability Matrix

Requirement	Related Use Case(s)
FR-001	UC-02 Post Surplus Food Batch
FR-002	UC-03 Browse Available Food; UC-04 Apply Dietary Filter
FR-003	UC-05 Reserve Food Batch
FR-004	UC-05 Reserve Food Batch «include» UC-06 Generate Pickup Verification Code
FR-005	UC-07 Verify Pickup
NFR-001	UC-08 Send Proximity Notification
NFR-002	UC-01 Register / Login; UC-05 Reserve Food Batch; UC-07 Verify Pickup

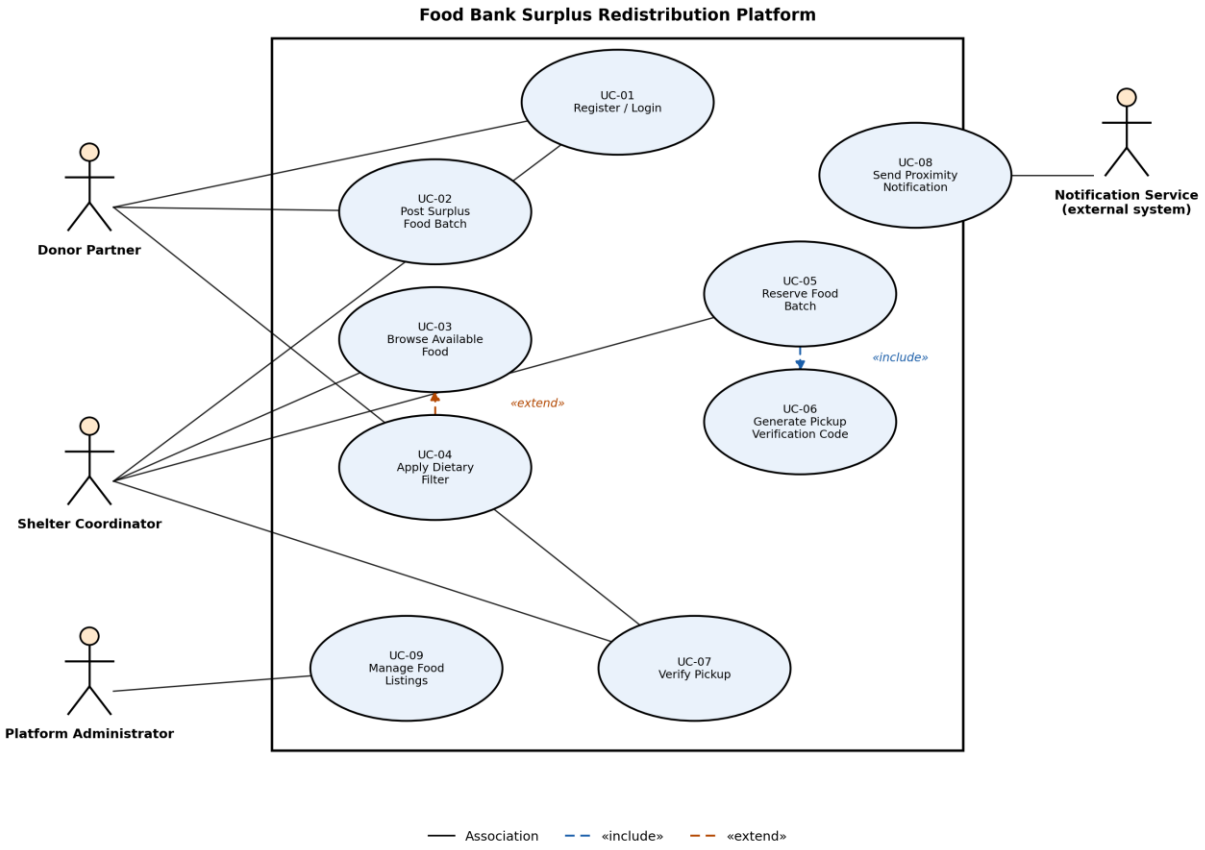
4. Use Cases

- UC-01 Register / Login — Any actor authenticates and, for Donor Partners and Shelter Coordinators, is verified before accessing platform features.
- UC-02 Post Surplus Food Batch — Donor Partner lists a surplus batch with quantity, dietary tags, and an expiry countdown timer.
- UC-03 Browse Available Food — Shelter Coordinator views all active, unexpired food batches.

- UC-04 Apply Dietary Filter — Shelter Coordinator optionally narrows the browse results by dietary tag, distance, or expiry window.
- UC-05 Reserve Food Batch — Shelter Coordinator claims an available batch for a specific pickup window.
- UC-06 Generate Pickup Verification Code — System creates a unique code tied to a confirmed reservation (always triggered by UC-05).
- UC-07 Verify Pickup — Donor Partner validates the Shelter Coordinator's code at handoff, completing the reservation.
- UC-08 Send Proximity Notification — Notification Service pushes an alert to shelters within 5 km when a high-quantity batch is posted.
- UC-09 Manage Food Listings — Platform Administrator reviews, edits, or removes listings to maintain platform integrity.

5. UML Use-Case Diagram

The diagram below models the Food Bank Surplus Redistribution Platform, with four actors, nine use cases, a labelled system boundary, plain associations, one «include» relationship, and one «extend» relationship.



PlantUML Source

```
@startuml
left to right direction
skinparam packageStyle rectangle
actor "Donor Partner" as Donor
actor "Shelter Coordinator" as Shelter
actor "Platform Administrator" as Admin
actor "Notification Service\n(external system)" as Notify

rectangle "Food Bank Surplus Redistribution Platform" {
usecase "UC-01\nRegister / Login" as UC1
usecase "UC-02\nPost Surplus Food Batch" as UC2
usecase "UC-03\nBrowse Available Food" as UC3
usecase "UC-04\nApply Dietary Filter" as UC4
usecase "UC-05\nReserve Food Batch" as UC5
```

```
usecase "UC-06\nGenerate Pickup Verification Code" as UC6
usecase "UC-07\nVerify Pickup" as UC7
usecase "UC-08\nSend Proximity Notification" as UC8
usecase "UC-09\nManage Food Listings" as UC9
}

Donor --> UC1
Donor --> UC2
Donor --> UC7
Shelter --> UC1
Shelter --> UC3
Shelter --> UC5
Shelter --> UC7
Admin --> UC9
Notify --> UC8

UC5 .> UC6 : <<include>>
UC4 .> UC3 : <<extend>>
@enduml
```

Textual Actor → Use-Case Representation

- Donor Partner → Register/Login, Post Surplus Food Batch, Verify Pickup
- Shelter Coordinator → Register/Login, Browse Available Food, Reserve Food Batch, Verify Pickup
- Platform Administrator → Manage Food Listings
- Notification Service (external) → Send Proximity Notification
- Reserve Food Batch «include» Generate Pickup Verification Code
- Browse Available Food ← «extend» — Apply Dietary Filter

6. Use-Case Flow Specification

Use Case ID: UC-05

Use Case Name: Reserve Food Batch

Primary Actor: Shelter Coordinator

Supporting Actors: None (system automatically triggers UC-06 Generate Pickup Verification Code)

Preconditions

- The Shelter Coordinator is authenticated and verified (UC-01 completed).
- The selected food batch is active and has not expired.
- The selected food batch has available quantity remaining.

Postconditions

- The food batch's available quantity is decremented and, if fully claimed, marked "Reserved."
- The chosen pickup window is recorded against the reservation.
- A unique pickup verification code has been generated and shown to the Shelter Coordinator (via UC-06).
- The reservation and updated batch status are persisted in the system.

Main Success Scenario

1. The Shelter Coordinator opens the Browse Available Food screen.
2. The system displays all active, unexpired surplus food batches.
3. The Shelter Coordinator selects a suitable food batch.
4. The system displays batch details, including expiry countdown, quantity, dietary tags, and available pickup windows.
5. The Shelter Coordinator selects an available pickup window and confirms the reservation request.
6. The system re-validates that the batch is still active and has available quantity.

7. The system reserves the batch and decrements its available quantity.
8. The system triggers UC-06 to generate a unique pickup verification code.
9. The system records the reservation, pickup window, and verification code together.
10. The system displays a confirmation screen with the pickup window and verification code to the Shelter Coordinator.
11. The use case ends successfully.

Alternate Flow

A1. Batch becomes unavailable during reservation (triggered at step 6):

A1.1. The system detects that the selected batch has expired or its available quantity has reached zero since it was last displayed.
A1.2. The system rejects the reservation and does not decrement quantity or generate a code.
A1.3. The system informs the Shelter Coordinator that the batch is no longer available.
A1.4. The system redirects the Shelter Coordinator to the Browse Available Food screen with updated results.
A1.5. The use case ends without a reservation being created.